

Chapter 5

Inverses and Radicals

A function sends each input to one output. Two questions about that sending decide whether it can be undone: does every input land on a distinct output, and does every possible output get hit? The first is the one-to-one property, the second is onto. When both hold the function has an inverse, a second function that runs the arrows backward, turning outputs into the inputs that produced them. Radicals are the inverses of power functions, and they carry the domain restrictions that the backward direction forces.

5.1 Domain and Range

The domain is the set of inputs a function accepts; the range is the set of outputs it produces. For a function given by a formula, the domain is every input for which the formula is defined, and the range is every value the formula attains as the input runs over the domain.

Definition: Domain and range. The *domain* of f is the set of all inputs x for which $f(x)$ is defined. The *range* is the set of all outputs $f(x)$ produced as x ranges over the domain.

Two operations restrict a domain. Division forbids a zero denominator, and an even root forbids a negative radicand. Everything else in this course accepts all real inputs.

Example 5.1.1. (*reading the domain from the formula*) For $f(x) = \frac{1}{x-5}$ the denominator vanishes at $x = 5$, so the domain is all reals except 5. For $g(x) = \sqrt{x-3}$ the radicand $x-3$ must be nonnegative, so the domain is $x \geq 3$. For $h(x) = x^2 + 1$ no operation restricts anything, so the domain is all reals, while the range is $y \geq 1$ because a square is never negative.

Common pitfall. Domain restrictions come from the operations, not from the graph's appearance. An odd root such as $\sqrt[3]{x}$ accepts negatives, since a cube root of a negative number is a real negative number; only *even* roots restrict the radicand.

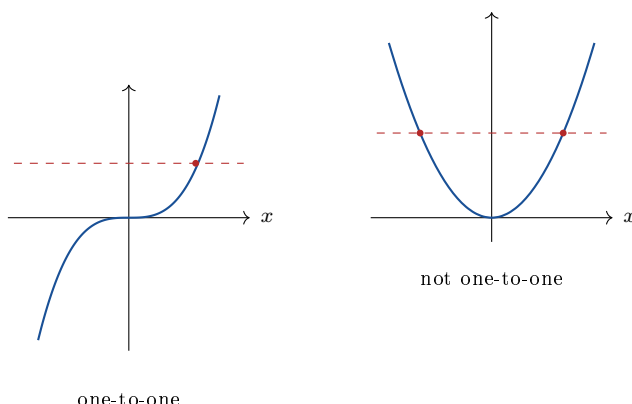
5.2 One-to-One and Onto

Two properties describe how a function distributes its inputs over its outputs.

Definition: One-to-one and onto. A function f is *one-to-one* if distinct inputs always give distinct outputs: $x_1 \neq x_2$ forces $f(x_1) \neq f(x_2)$. It is *onto* a target set of outputs if every value in that set is attained by some input.

One-to-one is the property that no output is shared by two inputs. It has a graphical test that mirrors the vertical line test for functions.

Key idea. Horizontal line test. A function is one-to-one if and only if no horizontal line crosses its graph more than once. A repeated crossing marks two inputs sharing one output, which violates the one-to-one property.



Example 5.2.1. (*applying the horizontal line test*) The line $f(x) = 2x - 1$ is one-to-one; every horizontal line meets it once. The parabola $f(x) = x^2$ is not; the horizontal line $y = 4$ meets it at $x = 2$ and $x = -2$, two inputs sharing the output 4. Restricting the parabola to $x \geq 0$ removes the left half and makes the restricted function one-to-one.

Whether a function is onto depends on the target set declared for its outputs. The squaring function is onto the nonnegative reals (every nonnegative number is a square) but not onto all reals (no input squares to a negative number). Choosing the target set to be exactly the range makes any function onto by construction.

5.3 The Inverse Function

When f is one-to-one, each output came from exactly one input, so the assignment can be reversed: send each output back to the input that produced it. That reverse assignment is the inverse function f^{-1} .

Definition: Inverse function. If f is one-to-one, its *inverse* f^{-1} is the function satisfying

$$f^{-1}(y) = x \quad \text{exactly when} \quad f(x) = y.$$

Equivalently, $f^{-1}(f(x)) = x$ for every x in the domain of f , and $f(f^{-1}(y)) = y$ for every y in the range.

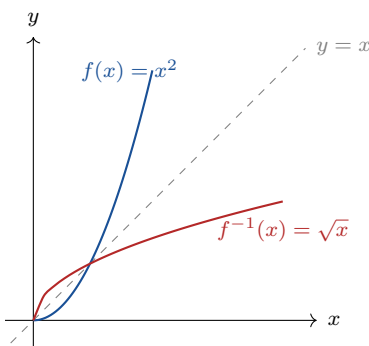
Key idea. An inverse undoes the function. Its domain is the range of f and its range is the domain of f ; the two sets trade places, because reversing the arrows swaps inputs and outputs. Only a one-to-one function has an inverse, since a shared output would have no single input to return to.

5.3.1 Finding an inverse

Because the inverse swaps the roles of input and output, the algebra is: write $y = f(x)$, exchange x and y , and solve for y . The result is $f^{-1}(x)$.

Example 5.3.1. (*finding an inverse*) Invert $f(x) = 2x - 1$. Write $y = 2x - 1$, swap to $x = 2y - 1$, and solve: $y = \frac{x+1}{2}$. So $f^{-1}(x) = \frac{x+1}{2}$. Check: $f^{-1}(f(x)) = \frac{(2x-1)+1}{2} = x$.

The picture. Swapping x and y reflects the graph across the line $y = x$. The graph of f^{-1} is the mirror image of the graph of f in that diagonal, because every point (a, b) on f becomes the point (b, a) on f^{-1} .



5.4 Radicals as Inverses of Powers

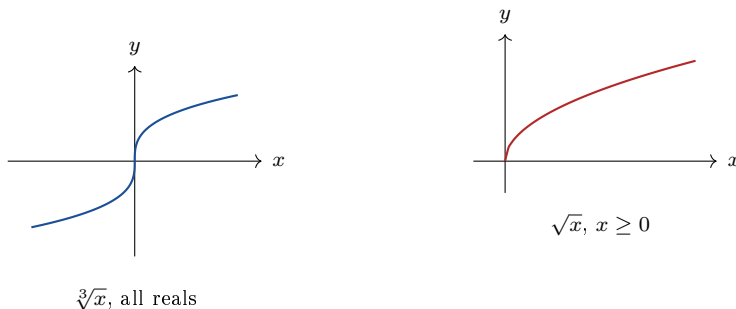
A radical is the inverse of a power function. The square root inverts squaring, the cube root inverts cubing, and in general $\sqrt[n]{x}$ inverts x^n . The parity of n controls the domains, exactly as it controlled the behavior of roots and ends in the polynomial chapter.

Key idea. Radicals and parity.

- n odd. x^n is one-to-one on all reals, so $\sqrt[n]{x}$ is defined for all reals; negatives are allowed.
- n even. x^n is not one-to-one on all reals (it shares outputs across $\pm x$), so it must be restricted to $x \geq 0$ before inverting. Its inverse $\sqrt[n]{x}$ then has domain and range $x \geq 0$.

Example 5.4.1. (*why the square root needs a restriction*) The function x^2 fails the horizontal line test on all reals. Restricted to $x \geq 0$ it is one-to-one, and its inverse is $\sqrt{x}$, with domain $x \geq 0$ and range $y \geq 0$. The discarded left half $x \leq 0$ would have inverted to $-\sqrt{x}$; the two halves are the two branches of the sideways parabola, only one of which is a function.

Example 5.4.2. (*an odd root accepts negatives*) The function x^3 is one-to-one on all reals, so $\sqrt[3]{x}$ is defined everywhere. For instance $\sqrt[3]{-8} = -2$, since $(-2)^3 = -8$. No restriction is needed because cubing never repeats an output.



5.4.1 Solving equations with radicals

A radical equation is solved by isolating the radical and raising both sides to the matching power. Raising an even root to an even power can introduce a false solution, because the squaring step accepts a sign the original equation forbade, so every candidate must be checked in the original equation.

Example 5.4.3. (*solving and checking*) Solve $\sqrt{x+5} = x-1$. Square both sides: $x+5 = x^2-2x+1$, so $x^2-3x-4=0$, giving $(x-4)(x+1)=0$ and candidates $x=4$ and $x=-1$. Check in the original: $x=4$ gives $\sqrt{9}=3$ and $4-1=3$, valid. $x=-1$ gives $\sqrt{4}=2$ but $-1-1=-2$; the two sides disagree, so $x=-1$ is rejected. The only solution is $x=4$.

Common pitfall. Squaring both sides can manufacture solutions the original equation never had, because $\sqrt{}$ returns the nonnegative root while squaring forgets the sign. Always substitute each candidate back into the original radical equation and discard any that fail.

5.5 Exercises

Exercises 5.5

- 5.5.1. State the domain of each: (a) $\frac{x}{x^2 - 9}$; (b) $\sqrt{2x - 6}$; (c) $\sqrt[3]{x + 1}$; (d) $\frac{1}{\sqrt{x - 4}}$.
- 5.5.2. Give the range of each, with a reason: (a) $f(x) = x^2 - 4$; (b) $f(x) = \sqrt{x} + 2$; (c) $f(x) = -|x|$.
- 5.5.3. Apply the horizontal line test to decide which are one-to-one: (a) $f(x) = 3x + 2$; (b) $f(x) = x^2 - 1$; (c) $f(x) = x^3$; (d) $f(x) = |x|$.
- 5.5.4. Explain, using the definition, why a function must be one-to-one to have an inverse. What goes wrong at a shared output?
- 5.5.5. Find the inverse of each and state its domain: (a) $f(x) = 3x - 7$; (b) $f(x) = \frac{x + 2}{5}$; (c) $f(x) = x^3 + 1$.
- 5.5.6. For $f(x) = x^2$ restricted to $x \geq 0$, find f^{-1} , verify $f^{-1}(f(x)) = x$ on the restricted domain, and state the domain and range of f^{-1} .
- 5.5.7. Explain why $\sqrt[3]{x}$ is defined for all real numbers while $\sqrt{x}$ is not, in terms of the parity of the power being inverted.
- 5.5.8. Sketch $f(x) = x^2$ on $x \geq 0$ and its inverse on the same axes, and show that each is the reflection of the other across $y = x$.
- 5.5.9. Solve and check, discarding any false solutions: (a) $\sqrt{2x + 3} = x$; (b) $\sqrt{x + 7} = x + 1$.
- 5.5.10. A function and its inverse meet the line $y = x$. Explain why any intersection of f with its inverse must lie on that line, using the reflection picture.
- 5.5.11. The function $f(x) = \sqrt{x - 2} + 1$ is built from the parent $\sqrt{x}$ by transformations. State the domain and range, find the inverse, and confirm the domain of the inverse equals the range of f .

5.5.12. Determine whether $f(x) = \frac{1}{x}$ is its own inverse by computing f^{-1} , and explain the symmetry of its graph in light of the result.